

A finite-codimensional obstruction for the mixed-weight heat-decay question of Bonet–Lusky–Taskinen

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Lane 11, model GPT5.6

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Status. Candidate counterexample / complete negative answer, likely valid, pending human review.

1 Source question

Bonet, Lusky, and Taskinen study Schauder bases whose finite-codimensional tails force arbitrarily fast decay for the heat equation [1]. Their Question 3° asks whether their higher-dimensional conclusion can hold when the weight is fast in only one coordinate:

$$w(x) = e^{|x_1|} \prod_{j=2}^N (1 + |x_j|)^2, \quad N > 1.$$

The question appears on page 14 of the source PDF, as equation (4.19).

2°. *Find a Schauder basis in the space $L_w^p(\mathbb{R}^N)$, $N > 1$, with the same properties as in Theorem 1.1* 1°.

Finally, we ask if it is possible in the case $N > 1$ to use the one-dimensional result in such a way that the weight only needs to be fast growing in one coordinate direction and milder assumptions are sufficient in other directions. For example:

3°. *Does Theorem 1.1, 2°, hold for the weight*

$$(4.19) \quad w(x) = e^{|x_1|} \prod_{j=2}^N (1 + |x_j|)^2.$$

Figure 1: The source discussion and Question 3° on page 14 of arXiv:1802.01902.

More explicitly, the requested conclusion would give a Schauder basis (e_n) of $L_w^1(\mathbb{R}^N)$ such that, for every $m \in \mathbb{N}$, some closed basis tail E_m satisfies

$$\|e^{t\Delta} f\|_\infty \leq C_m t^{-m} \|f\|_{1,w} \quad (f \in E_m, t \geq 1). \quad (1)$$

Every such E_m has finite codimension. We prove that no finite-codimensional subspace can have decay faster than $t^{-N/2-1}$ in operator norm.

2 Proof intuition

Let E have codimension d . At time t , place $d + 1$ copies of one fixed compactly supported bump along the x_2 -axis, with spacing $L\sqrt{t}$. Their supports are disjoint, and their span must meet E nontrivially. The mixed weight of the j th translated bump is only $O(j^2t)$, because the weight is quadratic in x_2 .

Normalize a nonzero combination in that intersection. At the center of the bump having the largest coefficient after weighted normalization, its own heat contribution is $\asymp t^{-N/2}$. All other contributions are smaller by fixed Gaussian factors $\exp(-cL^2|j - k|^2)$, so choosing L large prevents cancellation. Weighted normalization costs one further factor t^{-1} . This produces the unavoidable lower bound $t^{-N/2-1}$.

3 The obstruction theorem

Let

$$X = L_w^1(\mathbb{R}^N), \quad \|f\|_{1,w} = \int_{\mathbb{R}^N} |f(x)|w(x) dx,$$

with the mixed weight above. Write the heat kernel in the form

$$(H_t f)(x) = (e^{t\Delta} f)(x) = \gamma_N t^{-N/2} \int_{\mathbb{R}^N} \exp\left(-\frac{|x-y|^2}{4t}\right) f(y) dy,$$

where the positive normalization constant γ_N is immaterial.

Theorem 1. *Let $N > 1$ and let $E \subset X$ be a closed subspace of finite codimension. There are constants $c_E > 0$ and $T_E \geq 1$ such that*

$$\|H_t|_E\|_{X \rightarrow L^\infty(\mathbb{R}^N)} \geq c_E t^{-N/2-1} \quad (t \geq T_E).$$

Consequently, no Schauder basis of X has the arbitrary tail-decay property (1). Thus Question 3° has a negative answer.

Proof. Fix a nonnegative, nonzero function $\varphi \in C_c^\infty(\mathbb{R}^N)$ whose support lies in the ball $B(0, \rho)$. Let

$$d = \dim(X/E), \quad D = d + 1.$$

Choose a constant $L > 0$, to be fixed below, and for $t \geq 1$ define centers and weighted normalizing constants by

$$z_j(t) = jL\sqrt{t}e_2, \quad a_{j,t} = \int_{\mathbb{R}^N} \varphi(u)w(z_j(t) + u) du, \quad 1 \leq j \leq D.$$

Set

$$g_{j,t}(x) = \frac{\varphi(x - z_j(t))}{a_{j,t}}.$$

For all sufficiently large t , the supports of these D functions are pairwise disjoint, and $\|g_{j,t}\|_{1,w} = 1$.

The D -dimensional space spanned by the $g_{j,t}$ meets E nontrivially, because X/E has dimension $d < D$. Hence there are scalars $c_1, \dots, c_D$, not all zero, such that

$$f_t = \sum_{j=1}^D c_j g_{j,t} \in E.$$

After rescaling, assume $\sum_j |c_j| = 1$. Disjointness of the supports then gives the exact identity

$$\|f_t\|_{1,w} = \sum_{j=1}^D |c_j| = 1.$$

Put $q_j = |c_j|/a_{j,t}$ and choose k with $q_k = \max_j q_j$. On the fixed support of φ , all weight factors other than the x_2 factor are bounded above by constants independent of j and t , while

$$(1 + |jL\sqrt{t} + u_2|)^2 \leq C_L j^2 t \quad (t \geq 1, |u_2| \leq \rho).$$

It follows that

$$a_{j,t} \leq C j^2 t, \quad \sum_{j=1}^D a_{j,t} \leq C_E t. \quad (2)$$

Since $1 = \sum_j |c_j| = \sum_j q_j a_{j,t}$, we obtain

$$q_k \geq \frac{1}{\sum_{j=1}^D a_{j,t}} \geq C_E^{-1} t^{-1}. \quad (3)$$

We now evaluate $H_t f_t$ at $z_k(t)$. The diagonal integral satisfies

$$\begin{aligned} I_{k,t} &:= \gamma_N t^{-N/2} \int \exp\left(-\frac{|u|^2}{4t}\right) \varphi(u) du \\ &\geq \gamma_N e^{-\rho^2/4} \|\varphi\|_1 t^{-N/2} =: A_0 t^{-N/2}. \end{aligned} \quad (4)$$

For $j \neq k$ and all sufficiently large t ,

$$|z_k(t) - z_j(t) - u| \geq L|j - k|\sqrt{t} - \rho \geq \frac{L}{2}|j - k|\sqrt{t} \quad (u \in \text{supp } \varphi),$$

and therefore the corresponding off-diagonal integral is at most

$$I_{j,k,t} \leq \gamma_N \|\varphi\|_1 t^{-N/2} \exp\left(-\frac{L^2|j - k|^2}{16}\right). \quad (5)$$

Choose L once and for all so large that

$$2\gamma_N \|\varphi\|_1 \sum_{r=1}^{\infty} \exp(-L^2 r^2/16) \leq \frac{A_0}{2}.$$

Using $q_j \leq q_k$, the reverse triangle inequality together with (4)–(5) yields

$$\begin{aligned} |(H_t f_t)(z_k(t))| &\geq q_k I_{k,t} - \sum_{j \neq k} q_j I_{j,k,t} \\ &\geq \frac{A_0}{2} q_k t^{-N/2} \\ &\geq c_E t^{-N/2-1}, \end{aligned}$$

where the last line uses (3). Since $f_t \in E$ and $\|f_t\|_{1,w} = 1$, this proves the operator-norm lower bound.

Finally, suppose a Schauder basis had the property requested in the source. Choose an integer $m > N/2 + 1$ and let E_m be its corresponding closed basis tail. This tail has finite codimension, so the lower bound just proved and (1) would imply, for all sufficiently large t ,

$$c_{E_m} t^{-N/2-1} \leq C_m t^{-m}.$$

Multiplication by $t^{N/2+1}$ and passage to $t \rightarrow \infty$ gives $c_{E_m} \leq 0$, a contradiction. Hence the requested basis does not exist. $\square$

4 Verification notes and scope

The proof establishes more than failure of one tensor-product construction: it obstructs the desired estimate on every closed finite-codimensional subspace of $L_w^1(\mathbb{R}^N)$. The exponent $N/2 + 1$ comes from ordinary heat smoothing ($N/2$) plus the quadratic transverse weight (one further power of t at distance $\sqrt{t}$).

The following points were checked adversarially.

1. Translation occurs only in the second coordinate, which exists because $N > 1$; the exponential x_1 factor therefore remains uniformly bounded on the translated bump supports.
2. The vector in E may have arbitrary real or complex coefficients. Positivity of the Gaussian and the reverse triangle inequality make the diagonal-dominance estimate insensitive to their phases.
3. The test vector may depend on t , as is appropriate for a lower bound on the operator norm of $H_t|_E$.
4. Constants are allowed to depend on E (through $D = d + 1$), but never on t . This is enough to contradict any faster power bound.
5. No computational assertion or unproved auxiliary lemma enters the proof.

5 Novelty check and limitations

A bounded search on 9 August 2026 used arXiv, exact-title searches, the author names, the displayed weight, and the phrases “fast growing in one coordinate” and “Schauder basis heat equation.” It located the arXiv source and author-hosted versions, including a later manuscript that proves a stronger higher-dimensional theorem for weights fast in every direction. No paper or manuscript answering the displayed mixed-weight question was found. The later theorem does not apply here: the factors $(1 + |x_j|)^2$ fail the all-orders growth assumption in the transverse directions.

This is a bounded, not exhaustive, novelty review. Human review should focus on the normalization estimate (2)–(3) and the Gaussian diagonal-dominance step. Subject to that review, the argument is a complete negative answer to the literal Question 3°.

References

- [1] J. Bonet, W. Lusky, and J. Taskinen, *Schauder bases and the decay rate of the heat equation*, arXiv:1802.01902 [math.FA], 2018, Question 3°, p. 14.